Grapheme-colour synaesthesia may not improve performance in tests of spelling

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1. ABSTRACT

Grapheme-colour synaesthesia is a familial condition in which letters, numbers, days of the week and months of the year elicit colours overlaid on a printed page or in the mind's eye. The present study implemented novel methodologies to address present conflict in the literature concerning memory and visual search faculties (both involved in spelling), amongst an undergraduate population. Finding no significant differences between groups or significant correlations between synaesthetic colours and task performance, the results indicated that synaesthetic colours may not provide any significant advantage in spelling.

2. INTRODUCTION

In line with classic memory theory (Moscovitch & Craik, 1976; Paivio, 1969), grapheme-colour synaesthetes are thought to use their letter or word colour assignments to act as distinctive markers to differentiate target characters or to act as cues for later recall (Cytowic, 1989; Rich & Mattingley, 2003). While some studies have found a memory advantage amongst grapheme-colour synaesthetes (e.g. Mills, Innis, Westendorf, Owsianiecki, & McDonald, 2006), others reported scores within normal limits (Rothen & Meier, 2010) or no advantage at all (Rothen & Meier, 2009). Similarly, visual search studies reported performance advantages (e.g. Palmeri, Blake, Marois, Flanery, & Whetsell, 2002), or no difference to controls (e.g. Edquist, Rich, Brinkman, & Mattingley, , 2006).

3. AIMS & HYPOTHESES

- To address the ambiguity in the literature using novel methodology.
- Grapheme colour synaesthetes will respond faster and more accurately than control participants in all tasks.
- Synaesthetes will report using their synaesthetic colours to aid learning.

4. METHOD

PARTICIPANTS

Grapheme-colour synaesthetes (9): 6 female, 3 male; mean age: 19.9 yrs, SD: 1.9

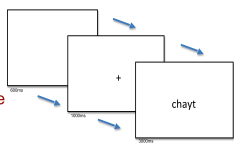
Controls (32):19 female, 13 male; mean age: 20.5 yrs, SD: 3.2

DESIGN

Quasi-experimental design. Between subjects analysis comparing scores of syneasthetes and controls. Within subjects design to assess the relationship between scores amongst synaesthetes.

MATERIALS

- Superlab 4.0 experiment presentations on 21.5" iMac desktop computers
- 'The Synesthesia Battery' (Eagleman, Kagan, Nelson, Sagaram, & Sarma, 2007), used via the Internet to assess for synaesthesia.



PROCEDURE

1) Learning Task: 27 novel non-words, three-word blocks. Word-stem completion task following each block.
2) Filler Task: Identification of two-digit numbers as higher or lower than the preceding number.
3) Word Recognition Task: Identification of previously learned non-words amongst incorrectly spelled non-words and unlearned non-words.
4) Letter Search Task: Search for target letter in common, English words.
5) Learning Method Questionnaire: Tick one or more of seven statements to indicate learning strategy used.

5. RESULTS

Table 1. Summary of **Mann-Whitney U** analyses of reaction times (ms) and accuracy scores (percentage of correct responses) for grapheme-colour synaesthetes and controls in the non-word recognition task and the letter search task. [NS: Not significant]

	U	z	p	Synaesthete Ranks (n = 9)	Control Ranks (n = 32)
Non-word recognition task					
(1) Reaction Times	128	-.5	NS	22.78	20.50
(2) Accuracy Scores	100	-1.4	NS	25.89	19.63
Letter Search Task					
(3) Reaction Times	90	-1.7	NS	15.00	22.69
(4) Accuracy Scores	115	-.94	NS	17.78	21.91

Table 2. **Pearson's r** analyses of the correlation between number of unique synaesthetic colours and reaction times (ms) and accuracy scores (percentage of correct responses) for grapheme-colour synaesthetes in the non-word recognition task and the letter search task. [NS: Not significant]

	r	n	p
Non-word recognition task			
(5) Unique Colours vs. Reaction Times	-.001	9	NS
(6) Unique Colours vs. Accuracy Scores	.277	9	NS
Letter Search Task			
(7) Unique Colours vs. Reaction Times	.009	9	NS
(8) Unique Colours vs. Accuracy Scores	.093	9	NS

A series of Mann Whitney U tests revealed no significant differences between grapheme-colour synaesthetes and controls for mean reaction times or accuracy percentage scores in either the word-recognition or letter search tasks (see Table 1 for a full outline of results).

Scatter-plots and a series of Pearson's product-moment correlations showed either a very weak or no relationship between the number of unique synaesthetic colours and task scores (see Table 2 for a full outline).

The majority of synaesthetes did not report the use of their synaesthetic colours as a memory aid during the learning task, only two noting the distinctiveness of individual letters or whole words.

6. DISCUSSION

Due to the lack of significance in the statistical analyses, the results indicate that synaesthesia may not have an impact on aspects of spelling, in line with the findings of Rothen and Meier (2009) and Edquist and colleagues (2006), among others.

A majority of 'associators' (colours in the 'mind's eye') rather than projectors (colours overlaid on text) in this study may be a reason for average performances (e.g. Palmeri et al., 2002). The questionnaire suggested that synaesthetes may need focused attention to elicit and benefit from their synaesthetic colours (Mattingley, Payne, & Rich, 2006; Rothen & Meier, 2009).

Future research may benefit from the use of a standardised spelling and word-recognition battery, a larger and more diverse sample of synaesthetes and direct instructions encouraging the use of synaesthetic abilities during participation to investigate colour-enhanced memory and search performance.

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